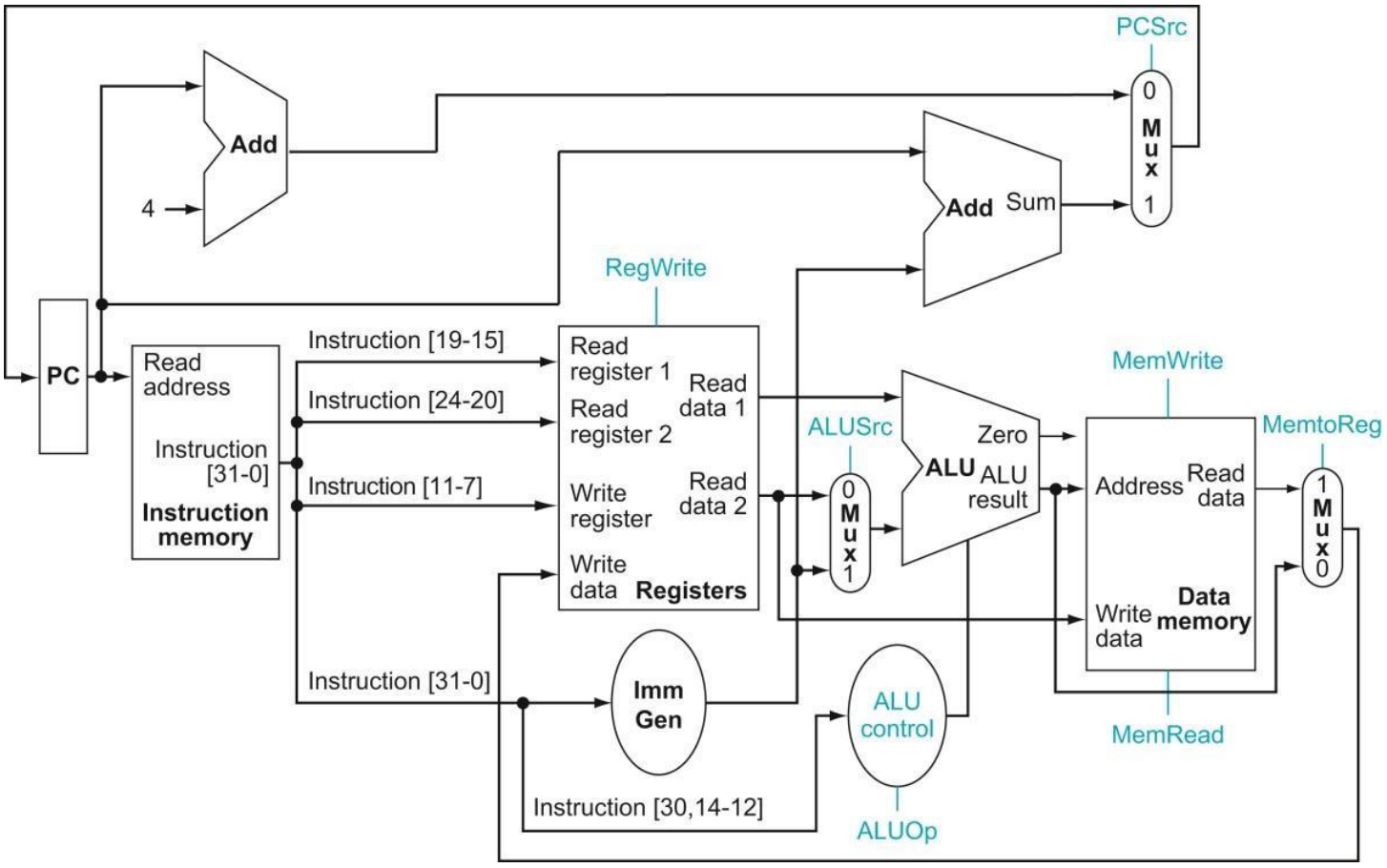
U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

Foundations of Computational Systems (FSC) - L.EIC004
Bachelor in Informatics and Computing Engineering
Department of Informatics Engineering (DEI)

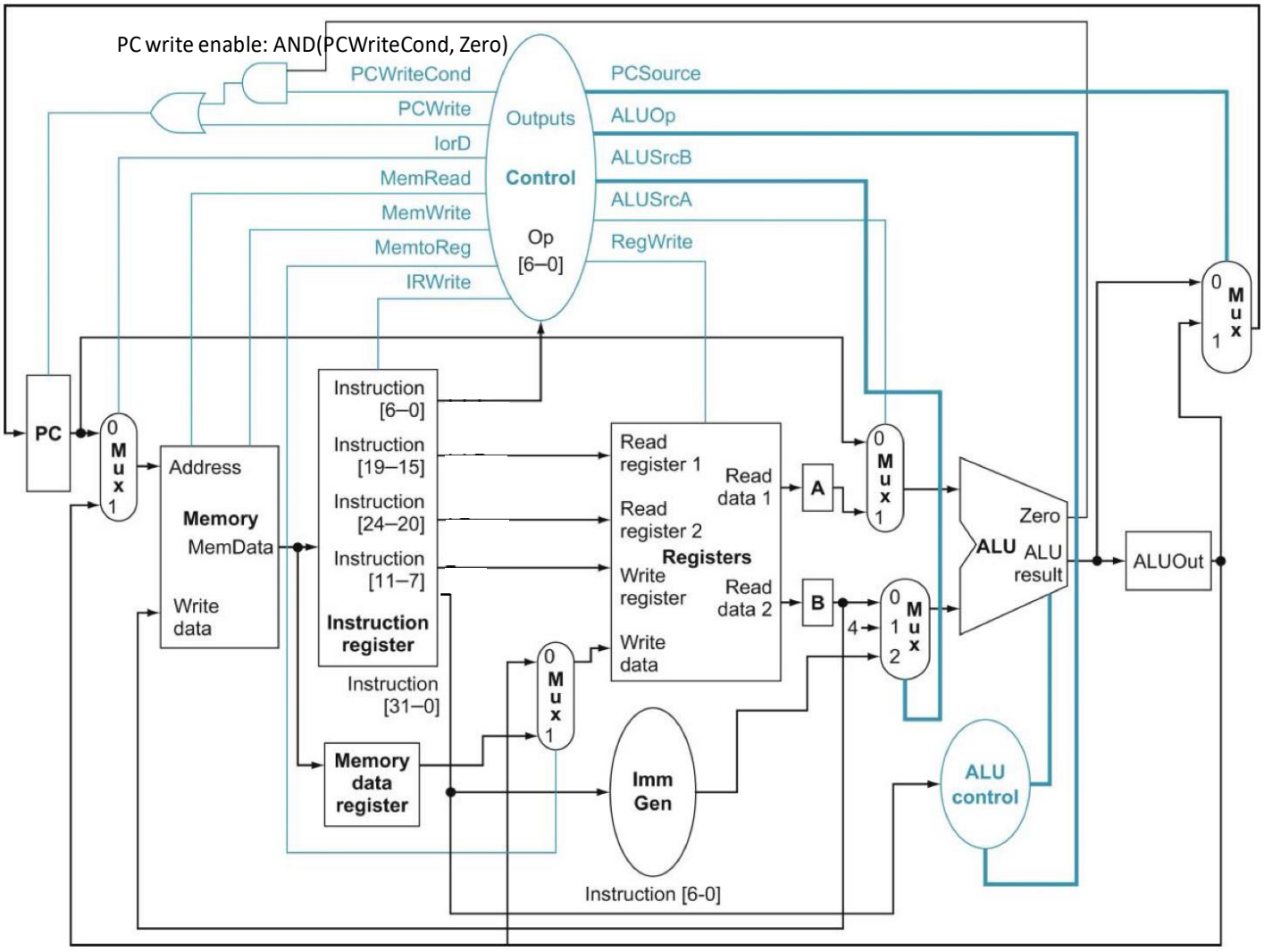
ALU control

Opcode	Operation	ALUOp	ALU operation
lw	load word	00	add
sw	store word	00	add
beq	branch if equal	01	subtract
R-type	Add	10	add
R-type	Sub	10	subtract
R-type	and	10	AND
R-type	Or	10	OR

Datapath of single-cycle CPU



Datapath of multi-cycle CPU



Operations for each clock cycle

Step name	Action for R-type instructions	Action for memory reference instructions	Action for branches
Instruction fetch	IR <= Memory[PC] PC <= PC + 4		
Instruction decode/register fetch	A<= Reg [IR[19:15]] B <= Reg [IR[24:20]] ALUOut <= PC + immediate		
Execution, address computation, branch/jump completion	ALUOut <= A op B	ALUOut <= A + immediate	if (A == B) PC <= ALUOut
Memory access or R-type completion	Reg [IR[11:7]] <= ALUOut	Load: MDR <= Memory[ALUOut] or Store: Memory [ALUOut] <= B	
Memory read completion		Load: Reg[IR[11:7]] <= MDR	

Diagram of finite-state machine control

